LionReel Studio

Autonomous Multi-Modal Video Slicing & Quality-Gated Viral Compilation Suite

FINAL PROJECT PROPOSAL

Project Title: LionReel Studio

Project Category: Capstone / Final Year Project

Primary Domain: Computer Vision & Media Engineering

Target Medium: YouTube Shorts, TikTok, Instagram Reels (9:16)

Author / Candidate: [Student Name / Registration Number]

Department: Dept. of Computer Science & Software Engineering

1. Executive Summary & Abstract

In modern digital broadcasting, short-form vertical video (YouTube Shorts, TikTok, Instagram Reels) is the fastest-growing medium of global media consumption. However, human curation, quality control, cropping, audio rebalancing, and compiling long-form video into high-retention 9:16 vertical shorts is labor-intensive and expensive.

LionReel Studio is an autonomous, end-to-end video curation and compilation system. It replaces manual editing workflows with algorithmic computer vision, acoustic frequency analysis, and automated composition. The system queries video sources, enforces a **12-stage automated quality gate** (eliminating blurry, shaky, watermark-contaminated, AI-synthetic, cartoon, or promotional content), detects viral peak moments using optical burst motion and audio energy, centers subjects dynamically with MOG2 background subtraction and Haar cascades, and renders color-graded, captioned short-form reels with custom music and branding.

2. Problem Statement & Motivation

> **The Challenge in Digital Media Production**
> Content creators, marketing agencies, and media companies spend 3 to 5 hours producing a single 45-second compilation reel. Naive scraping or basic ffmpeg scripts fail because they ingest blurry video, unstable camera shake, competitor logos, and synthetic AI clips, while cutting off moving subjects during 9:16 cropping.

- **High Latency & Editorial Overhead:** Sifting through raw repositories to locate entertaining climaxes is time-consuming.
- **Severe Quality Contamination:** Web-sourced video frequently contains low resolution (<480p), heavy motion blur, or invasive watermark logos.
- **Framing & Aspect Ratio Bottlenecks:** Arbitrary center-cropping cuts off moving pets and human subjects.
- **Creator Burnout:** Consistency demands daily publishing, which solo creators cannot sustain manually.

3. Programming Languages & Technology Stack

LAYER	TECHNOLOGY / TOOL	ARCHITECTURAL PURPOSE & ROLE
Core Engine	**Python 3.10+**	Orchestrates processing pipeline, math arrays, and automation backend.
Computer Vision	**OpenCV (cv2) & NumPy**	Laplacian variance, optical flow shake tracking, MOG2 subtraction, Haar cascades, Canny edges.
Media Processing	**MoviePy & FFmpeg / FFprobe**	Video cutting, frame transformation, -14 LUFS loudness normalization, and H.264 rendering.
Asset Generation	**Pillow (PIL)**	Dynamic stroke/shadow hook captions, watermark compositing, and thumbnail generation.
Content Ingestion	**yt-dlp**	Automated search query resolution, metadata filtering, and high-speed stream extraction.
Studio Backend	**Python `http.server`**	Lightweight local REST API handling process dispatching, config sync, and live log streaming.
Frontend GUI	**JavaScript (ES6+ Vanilla)**	Reactive UI state management, real-time log polling, video player, and 5-step stepper.
Styling & UX	**Modern CSS3 (Vanilla)**	Emerald dark/light mode engine, glassmorphic layout cards, responsive controls, and animations.

4. System Architecture & Workflow Pipeline

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[Step 1: Intelligent Query Ingestion (60+ Phrases)] | ▼ [Step 2: Metadata & Channel Reputation Filter] ──▶ (Rejects ads, vlogs & duplicate IDs) | ▼ [Step 3: 12-Stage Computer Vision Quality Gate] ──▶ REJECTS: Blur, Shake, Watermarks, AI, Cartoons | ▼ [Step 4: Peak Viral Moment Cutting] ──────────▶ (Burst motion + acoustic energy calculation) | ▼ [Step 5: Smart Crop V2 (MOG2 + Haar Cascades)] ──▶ DYNAMIC 9:16 CENTERING (80px Safety Margin) | ▼ [Step 6: Post-Production & Sound Engineering] ──▶ (CLAHE Color Grade, -14 LUFS Audio, Hook Caption) | ▼ [Step 7: Energy Ramp Sequencing & MP4 Export] ──▶ 1080x1920 @ 30 FPS + Auto-Thumbnails + Metadata
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5. The 12-Stage Automated Quality Gate

Every candidate clip is inspected through an automated sensory matrix before admission:

1. Sharpness & Resolution Check
Calculates Laplacian variance on 16 sample frames. Rejects blurry frames or low resolution (<480p).

2. Camera Shake Detector
Tracks background motion along the outer 15% border. Excludes erratic handheld phone footage.

3. Audio Music Detector
Computes RMS coefficient of variation (CV = σ / μ). Discards flat loud music; keeps natural pet audio.

4. Watermark & Logo Scanner
Evaluates Canny edge density & brightness in all 4 corner regions to eliminate competitor watermarks.

5. AI / Synthetic Video Filter
Analyzes texture variance, high-frequency noise, and morphing artifacts to reject Sora, Kling & Runway clips.

6. 2D Cartoon & Animation Filter
Color histogram quantization and edge density profiling discard anime, drawings, and cartoons.

7. Ad & Promo Banner Filter
Differentiates normal 1-line subtitles (allowed) from full-screen commercial slides and meme text walls.

8. Animal Presence Verification
Employs Haar Cascades (frontal cat/dog face) and motion clustering to guarantee genuine pet visibility.

9. Duplicate & Collision Filter
Cross-references video IDs and titles against historical logs to ensure zero content repetition.

10. Strict 9:16 Aspect Enforcement
Standardizes all outputs into modern smartphone vertical dimensions (1080x1920 pixels).

11. Adaptive Color Grading
Performs CLAHE contrast enhancement in LAB space with brightness balancing to 118 mean value.

12. Background Music Integration
Fades and overlays curated ambient backing tracks under original audio without clipping.

6. Mathematical Formulations & Key Algorithms

A. Peak Viral Moment Scoring

$$Score(t) = w_1 \cdot M_burst(t) + w_2 \cdot M_total(t) + w_3 \cdot C_center(t) + w_4 \cdot A_audio(t) - w_5 \cdot P_text(t)$$

Where **M_burst** is rapid optical acceleration, **C_center** rewards motion within the central 22%–78% vertical crop corridor, **A_audio** is acoustic RMS energy, and **P_text** penalizes full-screen promotional text walls.

B. Smart Crop V2 (MOG2 Centering with Safety Bounds)

Standard video tools center around (W/2, H/2), cutting off subjects at frame borders. Smart Crop V2 calculates foreground subject centroids using MOG2 background subtraction and Haar cascade bounding boxes, clamping with an 80px buffer:

$$C_x = clamp(median(X_foreground), (W_crop / 2) + Margin, W_source - (W_crop / 2) - Margin)$$

C. Energy Ramp Sequencing

Viewer retention drops within 3 seconds if the opening clip is dull. LionReel Studio isolates the single highest-energy clip as the opening hook, and sorts subsequent clips in ascending order of motion energy ($E_1 < E_2 < ... < E_n$) to produce an accelerating viral crescendo.

7. Interactive Studio Web Interface

The system is accompanied by a modern, responsive web application served locally via Python:

- **Real-Time Pipeline Status & Log Streamer:** Live console logging with elapsed process timers, step indicators, and progress tracking.
- **5-Step Creation Workflow:** Guides the operator through (1) Project Home, (2) Source Wizard, (3) Clip Editor, (4) Render Pipeline, and (5) Finished Video Library.
- **Granular Quality Controls:** Enables creators to toggle any of the 12 verifications, adjust thresholds, and choose slicing modes (Fast Action, Cute Reaction, Balanced Flow).
- **Finished Reel Manager:** Built-in vertical video player, metadata copy-to-clipboard, file downloader, and one-click cleanup.

8. Academic Novelty & Final Project Expansion Ideas

To achieve the highest marks in an academic evaluation, the project offers excellent expansion opportunities:

AREA	PROPOSED ENHANCEMENT	ACADEMIC / TECHNICAL IMPACT
Deep Learning	Integrate YOLOv8-nano / RT-DETR	Replaces Haar cascades with real-time multi-class pet/human object detection & tracking.
Speech & NLP	OpenAI Whisper Integration	Automated speech-to-text generating animated, word-by-word karaoke-style captions.
Multimodal LLM	Google Gemini 1.5 Flash API	Evaluates clip humor from keyframes; automatically crafts viral titles, hashtags, and SEO tags.
Social Automation	YouTube Data API & TikTok API	Provides 1-click scheduled publishing directly from the web dashboard.
Cloud Deployment	Containerized Cloud Run / AWS ECS	Transforms the tool from a desktop utility into a scalable multi-tenant SaaS.

9. Viva / Defense Key Talking Points

Examiner Presentation Strategy

When presenting to your panel or supervisor, highlight that this is not just a script—it is a **complete multi-modal engineering system** solving real-world digital media bottlenecks:

- **Multi-Modal Sensor Synthesis:** Integrates visual frame variance (Laplacian blur test), spatial temporal movement (Optical Flow shake test), and acoustic RMS variance.
- **High Computational Efficiency:** Relies on optimized classical CV algorithms that execute fast on standard CPUs without requiring expensive GPU servers.
- **Robust Software Architecture:** Decoupled architecture separating the algorithmic video worker, local REST API, and modern responsive frontend.